Adroit Technologies Protocol Driver

Name	ISOIL Email IMAP Client
DLL	ISOIL

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Contents

1. Overview

This driver is designed to provide an interface for ISOIL ISOMAG flow meters.

The data is sent in the form of an email containing a csv attachment.

Once the email is received it is decoded to expose the values contained in the attachment and body of the mail.

Inbox management is provided through a BACKUPBOX mailbox and an UNKNOWN/TODO box, where email messages can be moved as appropriate.

Messages older than a configured amount of days may be set for auto deletion.

A separate Adroit device has to be created for each flow meter with the station ID being set equal to the serial number of the flow meter.

There is a resource ID allocated automatically on configuration, new devices will attempt to find a compatible configuration (those that share an inbox for example) any others will create a new resource, selecting a resource in the dialogue will load the current configuration for that box, if the device is saved (on OK) any changes will be save with an appropriate warning. Resource ID can be deleted with appropriate warnings if it might affect other devices.

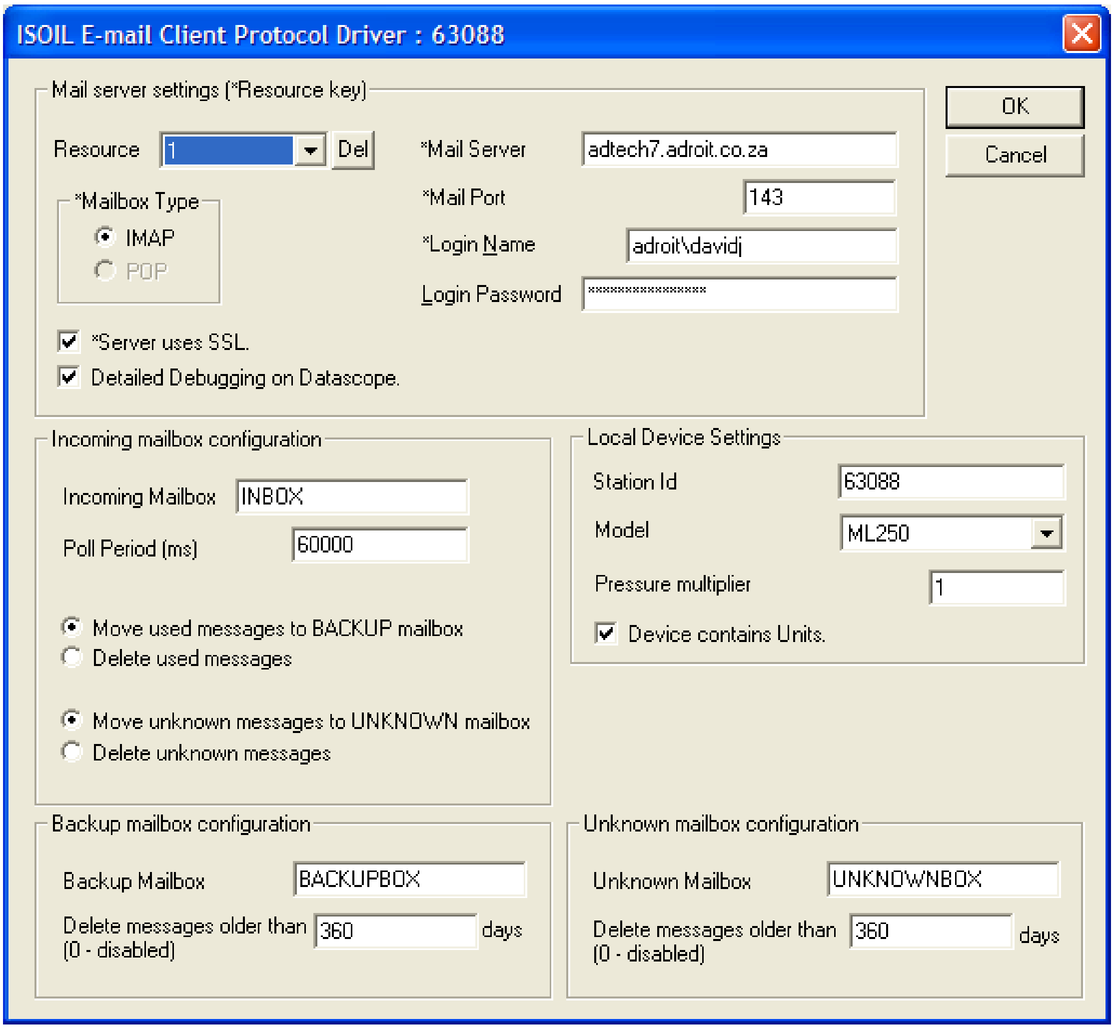
Offline devices and unknown (unconfigured) devices will have their emails moved to the UNKNOWN box if the MOVE radio is applied. Else if DELETE was selected unconfigured emails will be deleted. And offline device emails will be left in the main INBOX. Once an offline device is restarted the UNKNOWN box will be checked once for any emails for that device(s). This will also be done at Agent Server startup.

2. Wiring

Standard Ethernet network.

3. Device Configuration

3.1. Device Configuration



3.1.1. Mail Server Settings (*Resource key)

PLEASE NOTE

The parameters identified by the * define a resource common to all devices that share those parameters, any change to any other parameter in this group will be changed for all devices using this mailbox.

***Mailbox Type** – Choose IMAP or POP depending on your server type (**POP is no longer supported**).

***Mail Server** – The name of the mail server

***Mail Port** – 993 for IMAP (**143 is now no longer used**), 110 for POP (**no longer supported**).

***Login Name** – Username on the server.

***Server uses SSL** – (Default = ON) Some POP and IMAP servers require SLL mode. IMAP servers listen on port 143 typically, while **993** is used for SSL/TLS.
POP servers listen on port 110 typically, while **995** is used for SSL/TLS (**no longer supported**).

	PLEASE NOTE
The remaining parameters can be considered as sub-parameters (these will not result in a new resource being generated if different from any other resource) and if a change is requested on these the whole resource will be updated, and thus affect all devices using that resource.	

Login Password – User password.

Detailed debugging to datascope – Additional processing information will be sent to the datascope.

3.1.2. Incoming Mailbox Configuration

Incoming Mailbox – Incoming account inbox name. This is where all new emails should arrive first.

Poll Period (ms) – Mailbox will be checked for new mail at this interval.

Move used messages to BACKUP mailbox - If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to move that mail to the used or BACKUP mailbox, where further processing could take place such as remain there for x amount of days before being discarded (see further down).

Delete used messages – If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to delete the mail immediately.

Move unknown messages to UNKNOWN mailbox – If a mail is parsed and found to be junk, or for an offline device and NOT considered complete, checking this radio button will tell the driver to move that mail to the TODO or UNKNOWN mailbox, where further processing could take place such as remain there for x amount of days before being discarded (see further down).

Delete unknown messages – If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to delete the mail immediately.

3.1.3. Backup mailbox configuration

Backup Mailbox – Backup name – this is a sub or secondary mailbox where completed emails can be stored for a period of time.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

3.1.4. Local Device Settings

PLEASE NOTE
These setting are stored per device and will not affect other devices using the same resource ID.

Station ID – This is the identifier code sent in the subject line of the email For example, data from ML250 29902, the ID is then 29902.

Model – Select the Model of the ISOMAG flowmeter. Currently two models are supported. The ML150 and the ML250.
The ML155 and ML255 can be selected but these must run in compatibility mode.

Pressure Multiplier – Multiplier for the pressure values contained in the csv files

Device contains units – The ISOMAG can report data with unit fields.

3.1.5. Unknown mailbox configuration

Unknown Mailbox – Unknown name – this is a sub or secondary mailbox where emails not associated with an Adroit Device can be stored for a period of time. Junk/spam and or unconfigured or offline devices will also go here.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

4. Supported Addresses

4.1.1. ML150

A CSV row constitutes 1 record:
1,2010/02/03,00:00:00,+,<Unit 1>,<Param1>,<Unit 2>,<Param2>

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
Param 1	Positive volume counter	V1	V1	
Param 2	Flow rate	V3	V3	
Unit 1	Unit of Positive volume counter			U1
Unit 2	Unit of Flow rate			U3

4.1.2. ML250 option 1

A CSV row constitutes 1 record:
1,2010/02/03,00:00:00,+,<Unit 1>,<Param1>,-,<Unit 2>,<Param2>,<Unit 3>,<Param3>,<Unit 4>,<Param4>

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
Param 1	Positive volume counter	V1	V1	
Param 2	Negative volume counter	V2	V2	
Param 3	Flow Rate	V3	V3	
Param 4	Pressure	V4	V4	
Unit 1	Unit of Positive volume counter			U1
Unit 2	Unit of Negative volume counter			U2
Unit 3	Unit of Flow Rate			U3
Unit 4	Unit of Pressure			U4

4.1.3. ML250 option 2

A CSV row constitutes 1 record:
1,2010/02/03,00:00:00,+,<Unit 1>,<Param1>,-,<Unit 2>,<Param2>,<Unit 3>,<Param3>

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
Param 1	Positive volume counter	V1	V1	
Param 2	Positive volume counter	V2	V2	
Param 3	Flow Rate	V3	V3	
Unit 1	Unit of Positive volume counter			U1
Unit 2	Unit of Negative volume counter			U2
Unit 3	Unit of Flow Rate			U3

4.1.4. Universal Addressing

The following information is available for all flow meters

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
B1	Battery percentage 1	V5	V5	
B2	Battery percentage 2	V6	V6	
B3	Battery percentage 3	V7	V7	
Actual power source	Bitfield with: Bit 1 = ext Bit 2 = B1 Bit 3 = B2 Bit 4 = B3	V8	V8	
Antenna Signal	Antenna signal strength	V9	V9	

5. Protocol Definition and Considerations

5.1. Application Example

1. You have a ML250 ISOMAG flow meter from which you need to extract the following data:
    a. The positive volume counter
    b. The negative volume counter
    c. The Flow rate
    d. The Pressure

2. You are also required to extract the units of these values from the csv file.

3. In addition to this you need to determine the signal strength of the antenna.

In order to accomplish this you will need to create 5 Analog Agents and 4 String Agents.
You will scan the value of the positive volume counter (analog) from the "V1" address of the device.
You will scan the unit of the positive volume counter (string) from the "U1" address of the device.
You will scan the value of the negative volume counter (analog) from the "V2" address of the device.
You will scan the unit of the negative volume counter (string) from the "U2" address of the device.
You will scan the value of the flow rate (analog) from the "V3" address of the device.
You will scan the unit of the flow rate (string) from the "U3" address of the device.
You will scan the value of the pressure (analog) from the "V4" address of the device.
You will scan the unit of the pressure (string) from the "U4" address of the device.
You will scan the value of the antenna signal strength from the "V9" address of the device.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

6.3. Troubleshooting

1. Ensure the connection settings to your mailbox is correct.
2. Ensure the serial number of the ISOMAG flow meter matches the Station ID as set up in the driver configuration.
3. Ensure the emails do have attachments.
4. Ensure that the data attachment is before the event attachment.
5. Ensure that there is no invalid data in the body of the email, or in the attachment.
6. Do not move emails between folders in excel. A Copy is required for the driver to operate to specification.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
1	5-Feb-2010	First Released Document.
	29 April 2011	Added support for multiple models of ISOMAGs.
	22 June 2011	Added information from the body of the message.
	23 June 2011	Added introduction, application example and fault finding
	06 Dec 2012	Rev27 – Resource handling in registry fixed, offline devices emails moved to unknown, unknown checked at device start.
30	06-Jul 2021	Added ML255 model